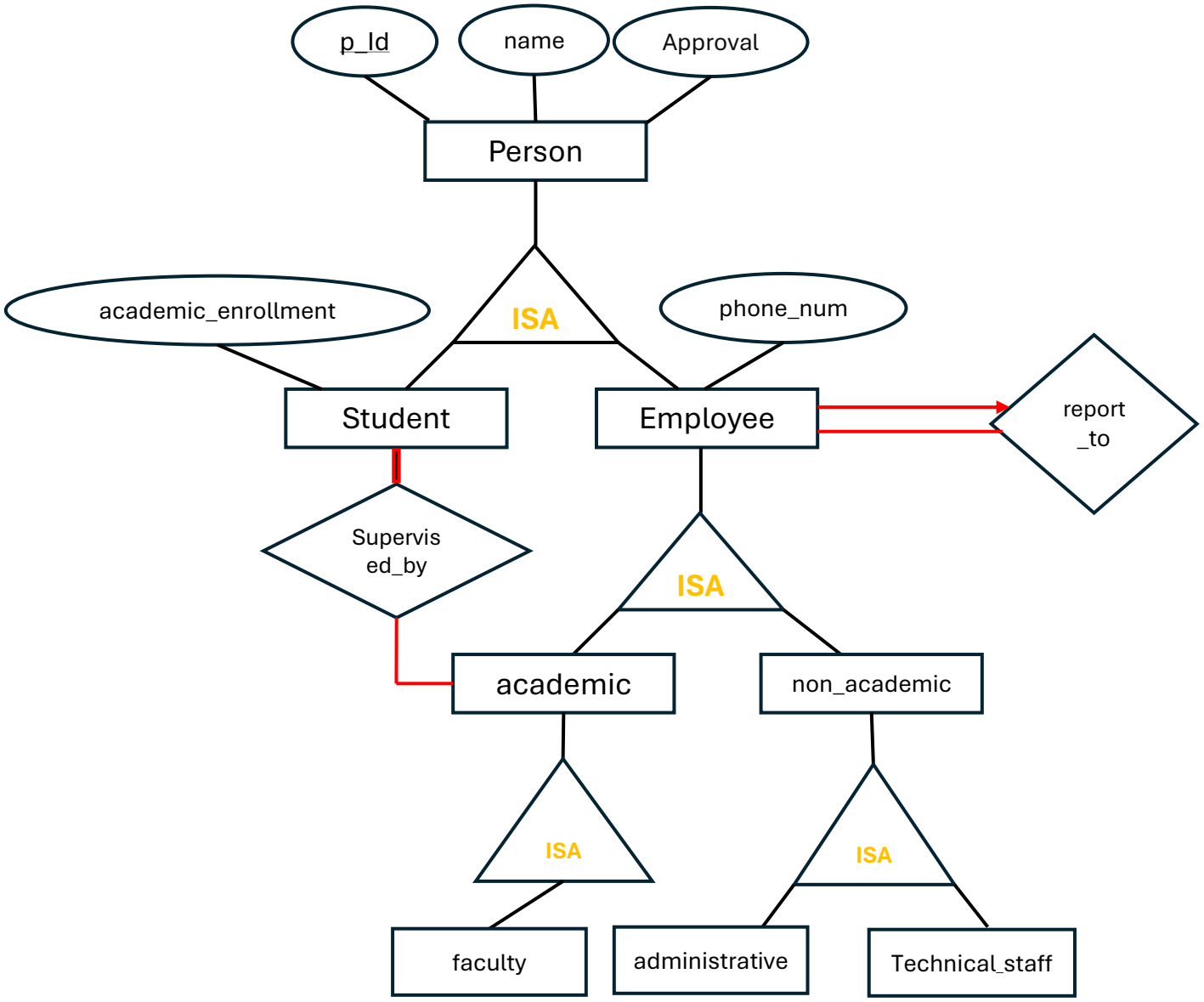
CONCEPTUAL DESIGN

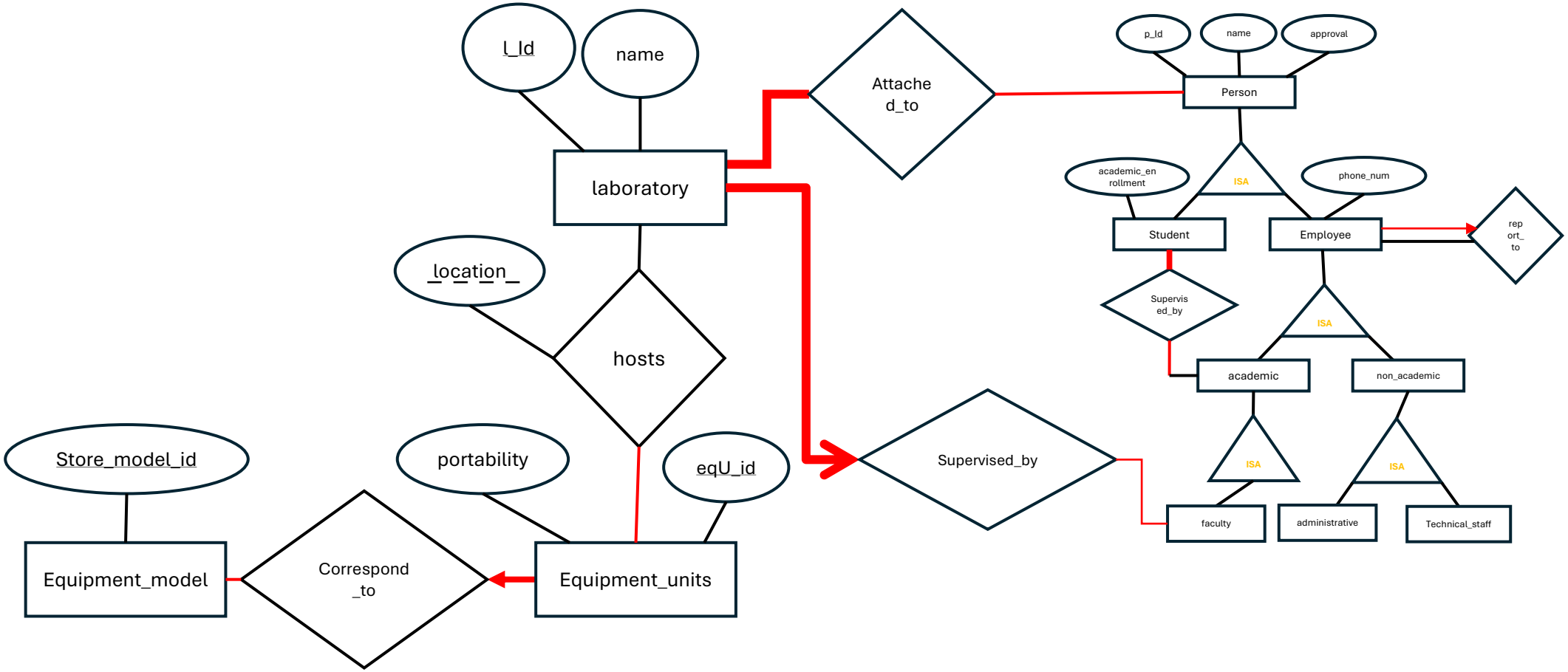
Research Lab Management System Project

As stated in the deliverables, the conceptual design is developed progressively. The initial slides present the main attributes, while subsequent versions progressively introduce additional relations and entities.

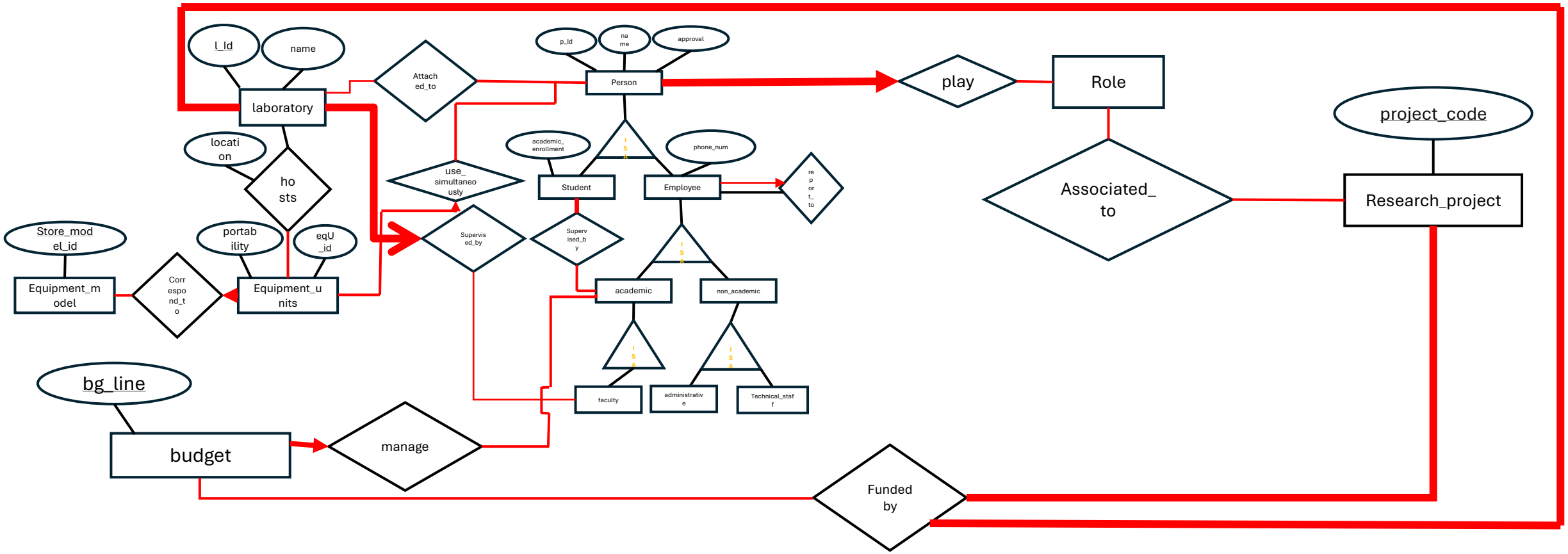
People


p_Id
name
Approval
Person
academic_enrollment
ISA
phone_num
Student
Employee
report
_to
Supervis
ed_by
ISA
academic
non_academic
ISA
ISA
faculty
administrative
Technical_staff

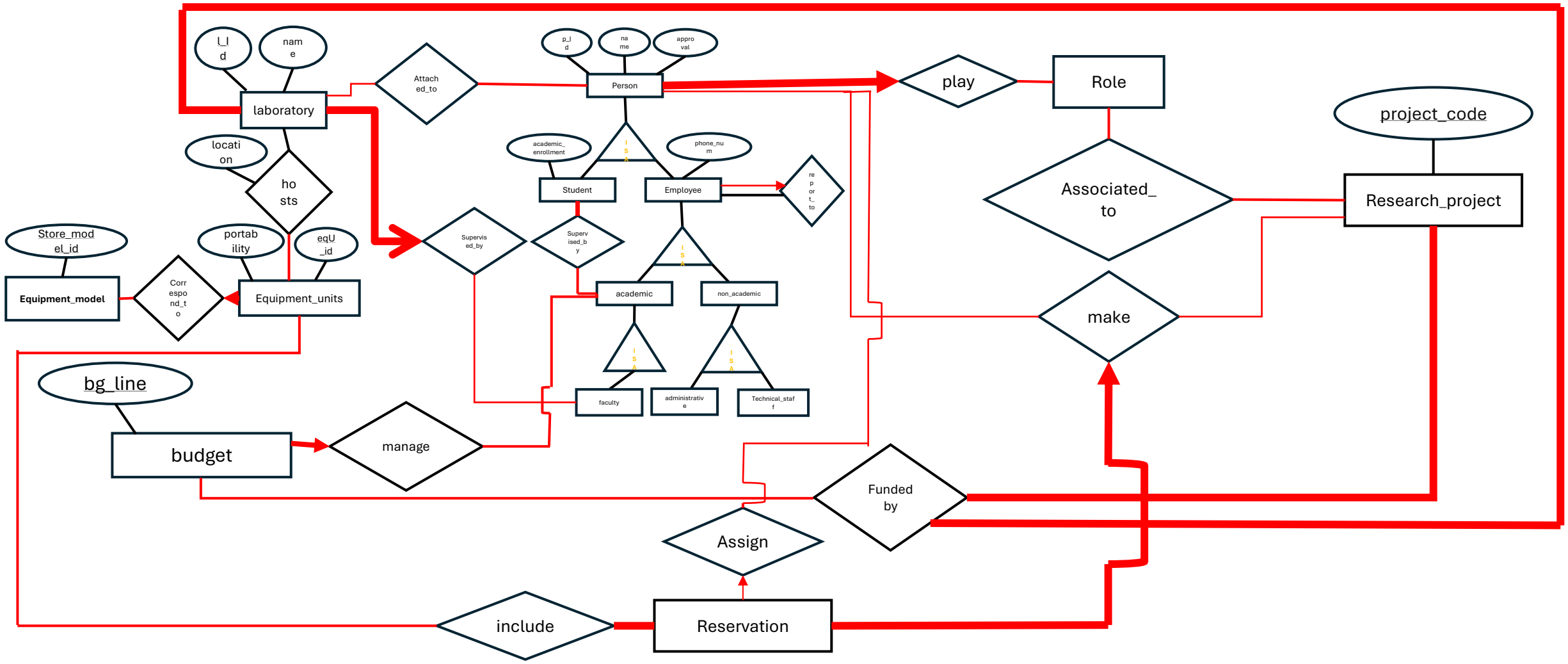
Laboratories, Equipments Units & models



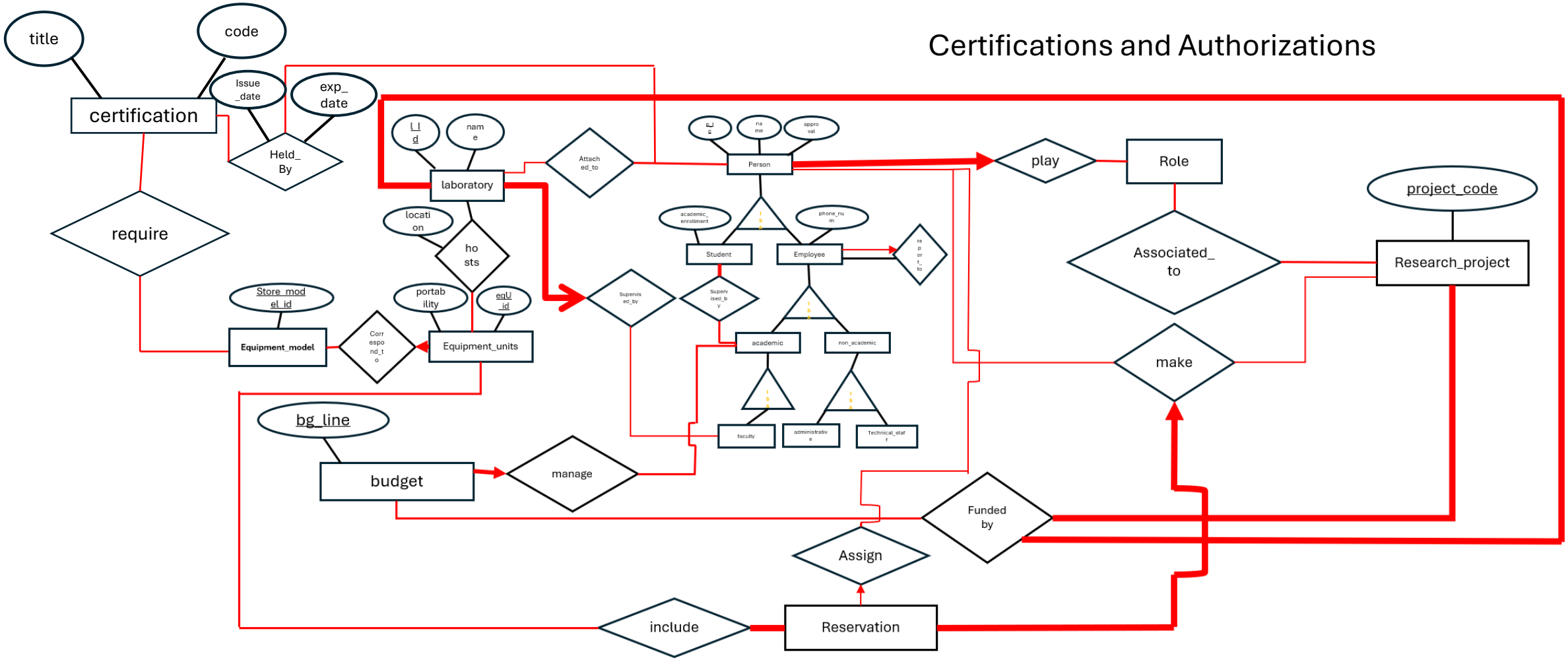
Research Projects, budgets



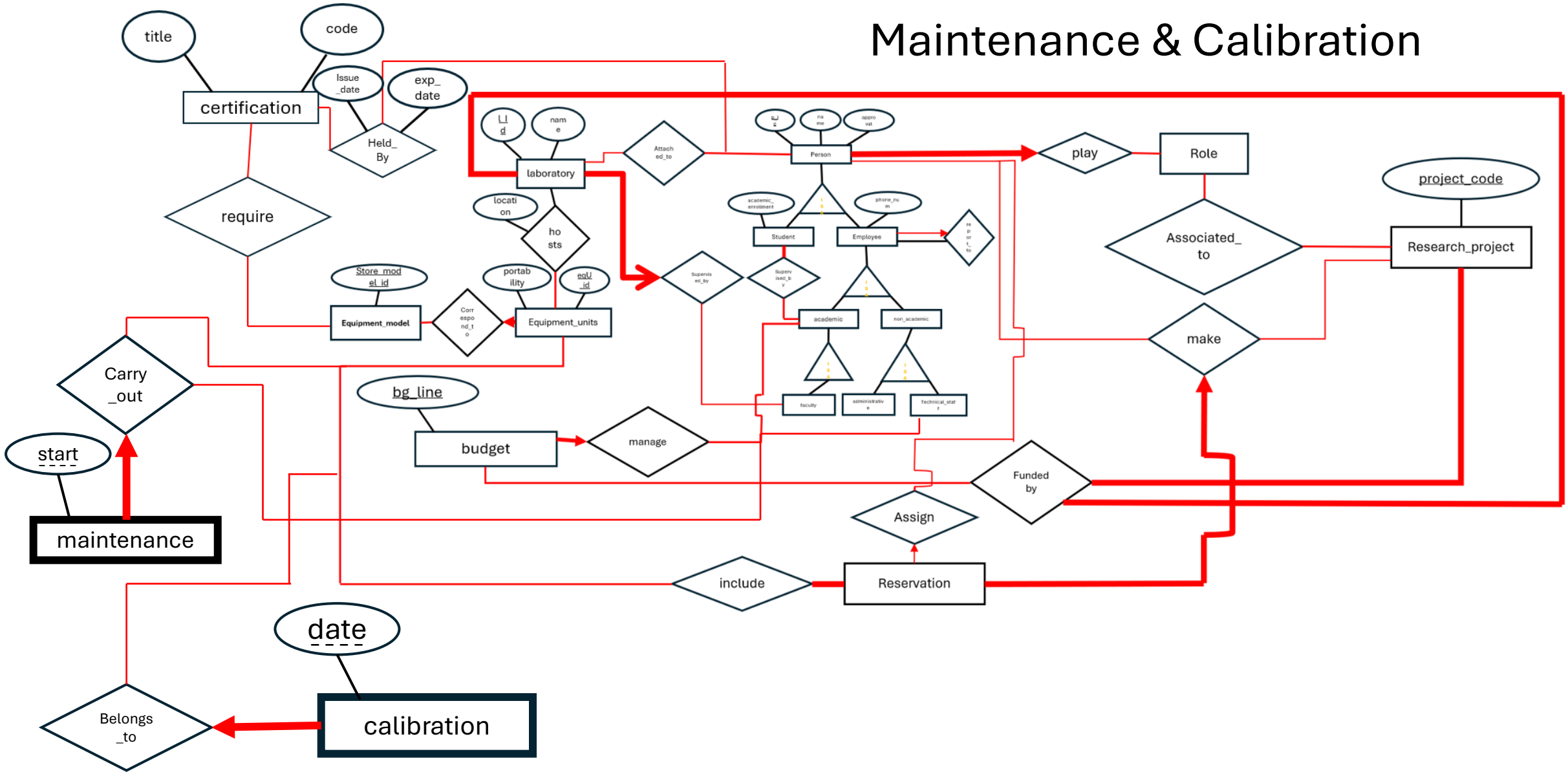
Reservations

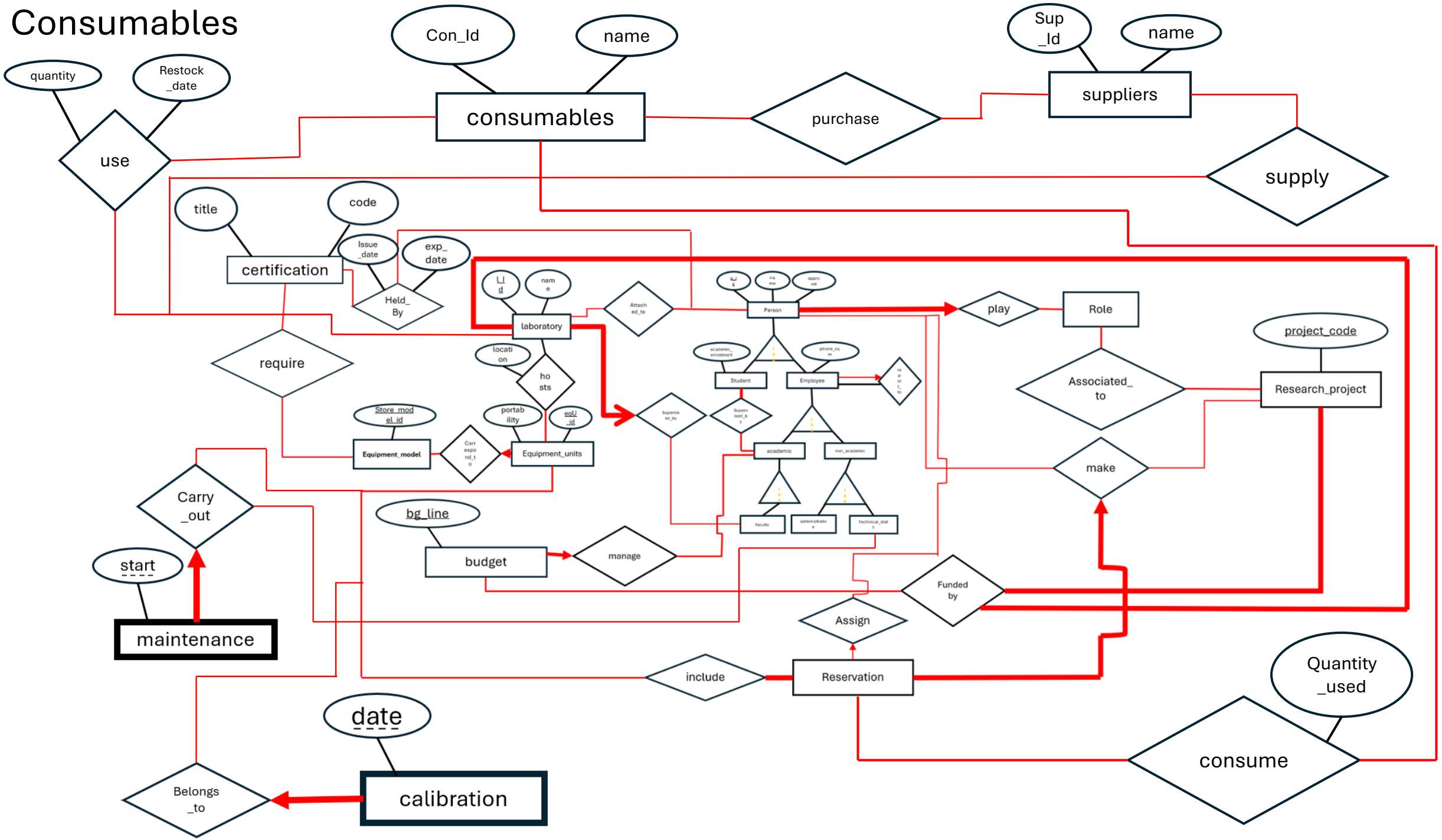


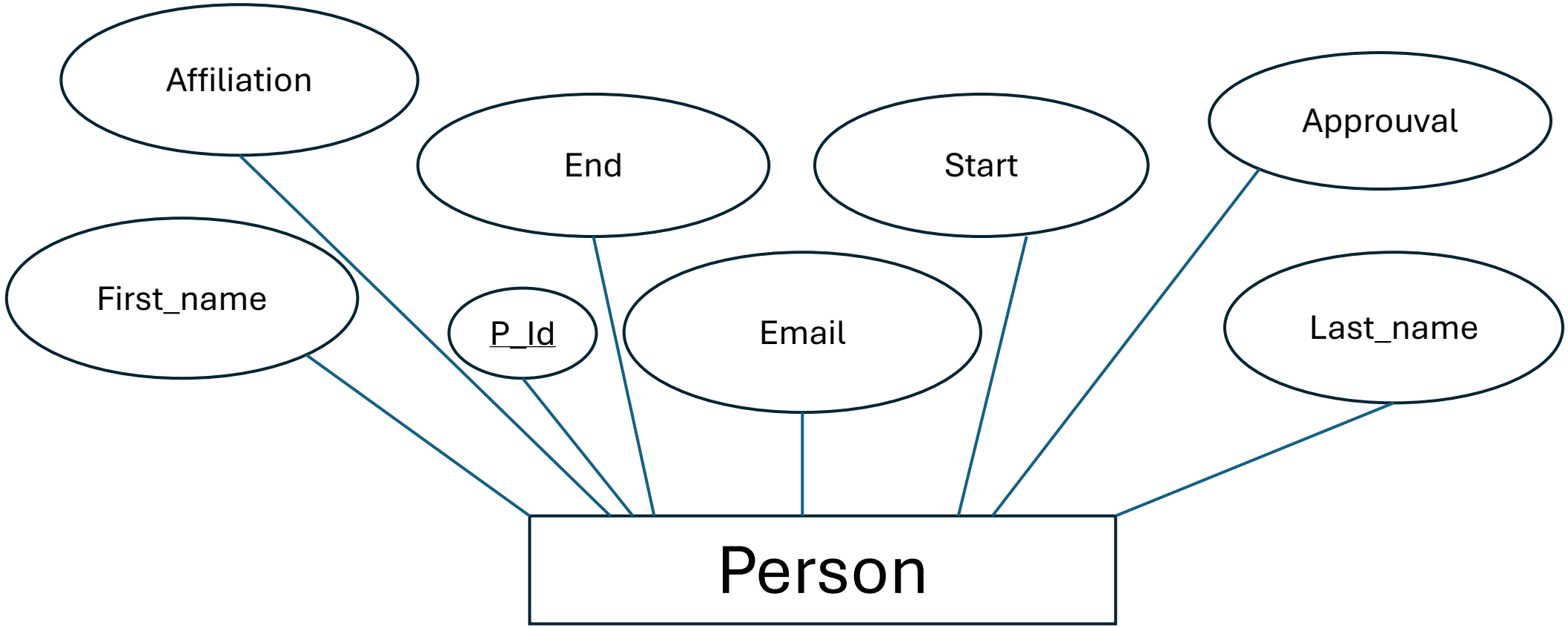
Certifications and Authorizations

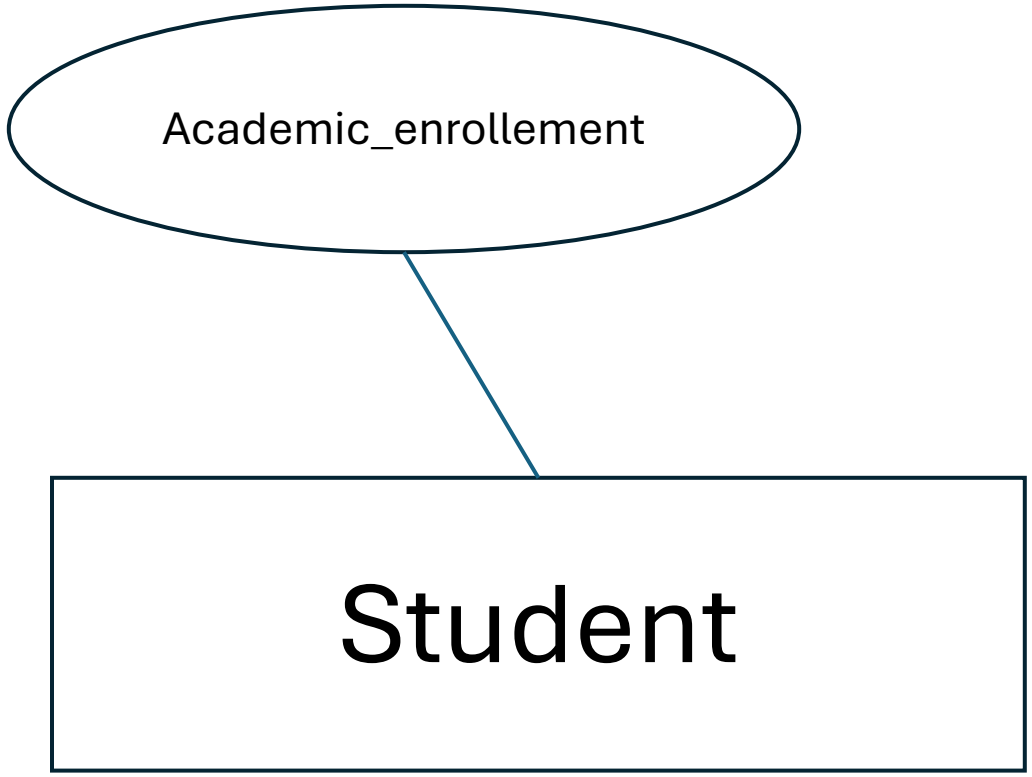


Maintenance & Calibration

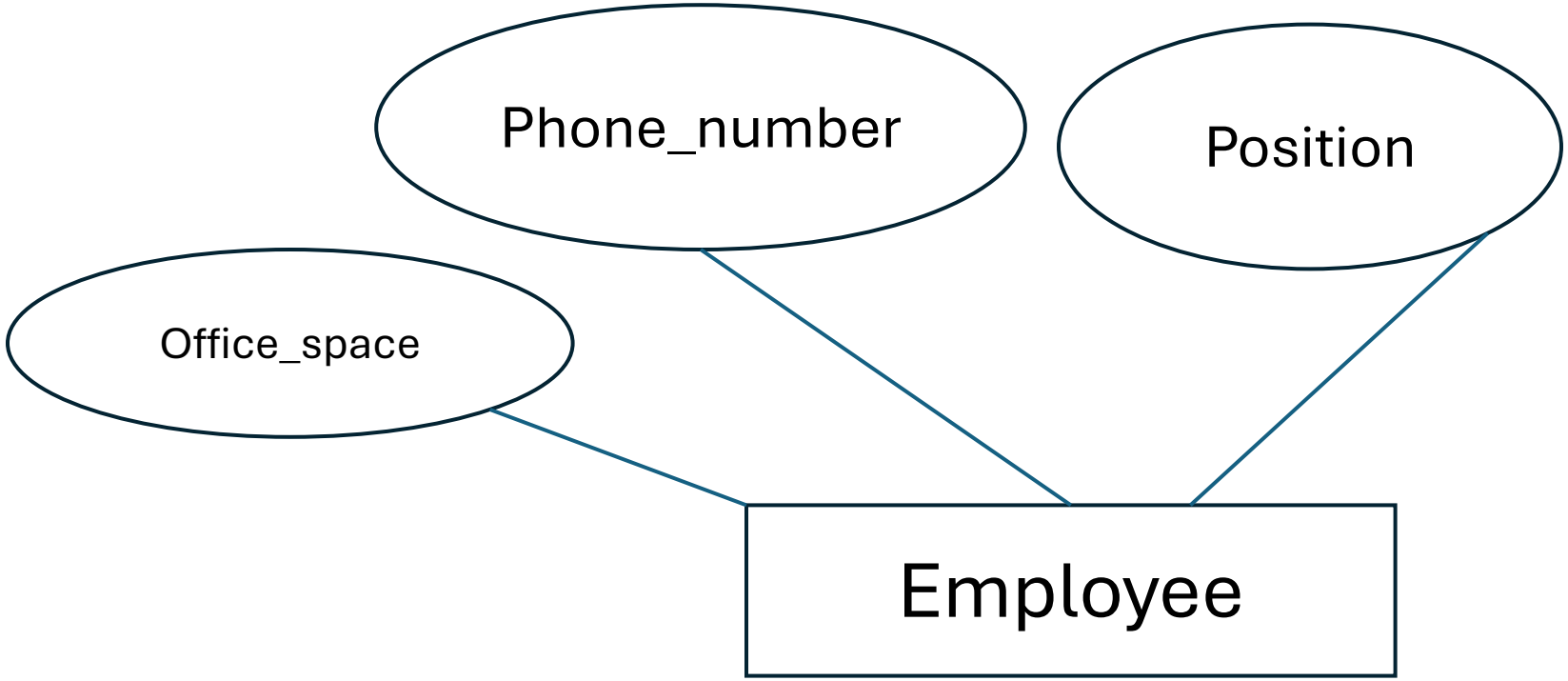



Consumables
Con_Id
name
quantity
Restock _date
use
consumables
purchase
Sup _Id
name
suppliers
supply
title
code
Issue _date
exp_ date
certification
Held_ By
require
laboratory
Attach ed_to
Person
play
Role
project_code
Associated_ to
Research_project
make
locati on
ho sts
Store_mod el_id
portab ility
equ _id
Equipment_model
Corr espo nd_t o
Equipment_units
Supervis ed_by
Student
Employee
academic
non_academic
faculty
administrativ e
technical_staf f
Carry _out
start
maintenance
bg_line
budget
manage
Funded by
Assign
include
Reservation
Quantity _used
consume
date
Belongs _to
calibration

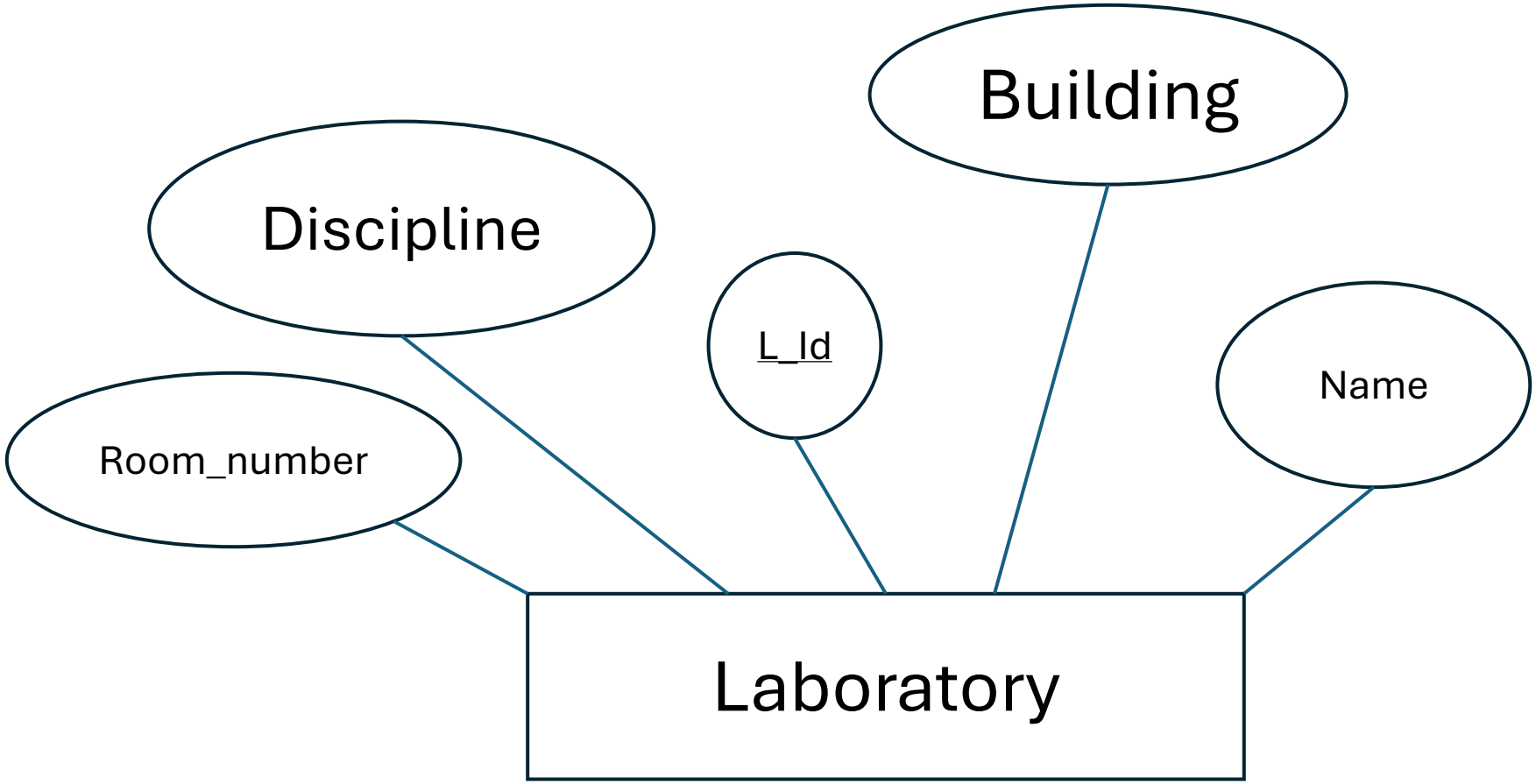

Affiliation
End
Start
Approuval
First_name
P_Id
Email
Last_name
Person

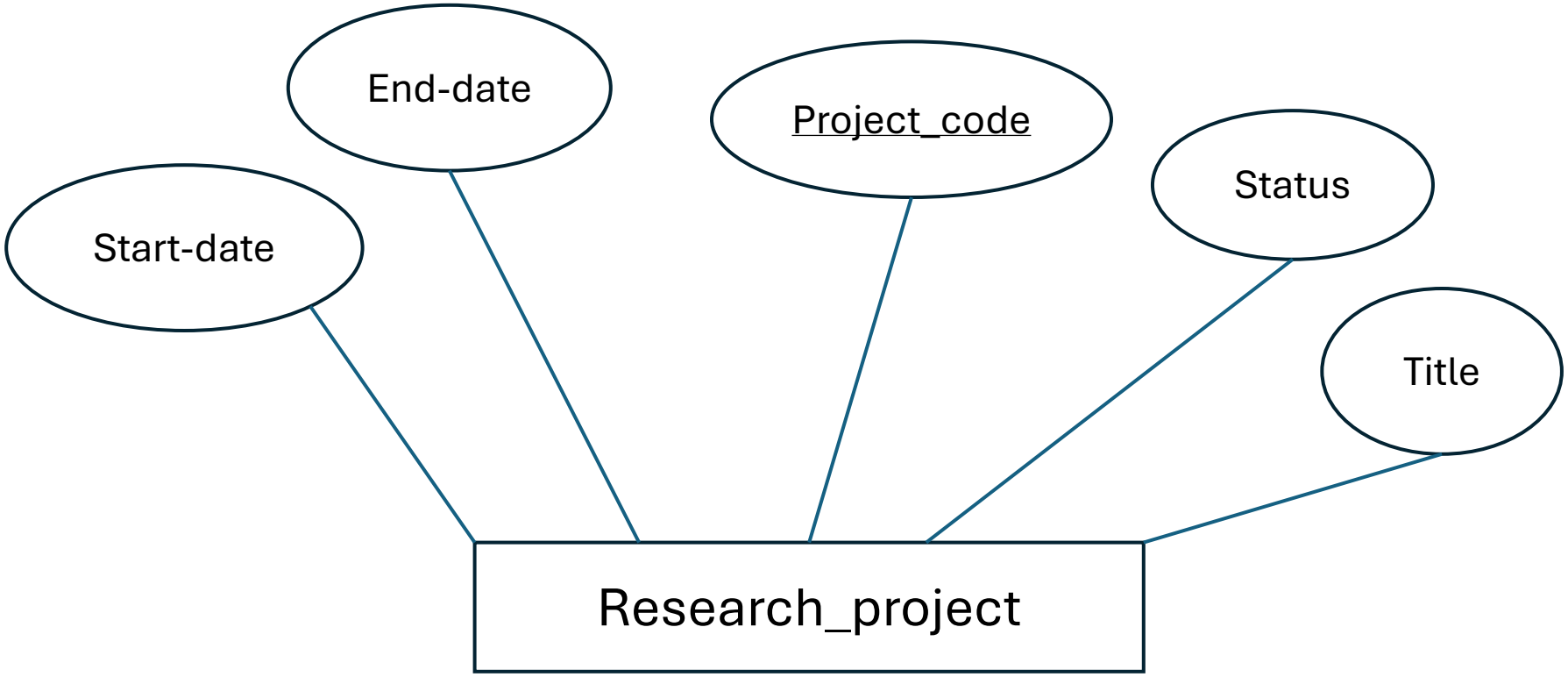


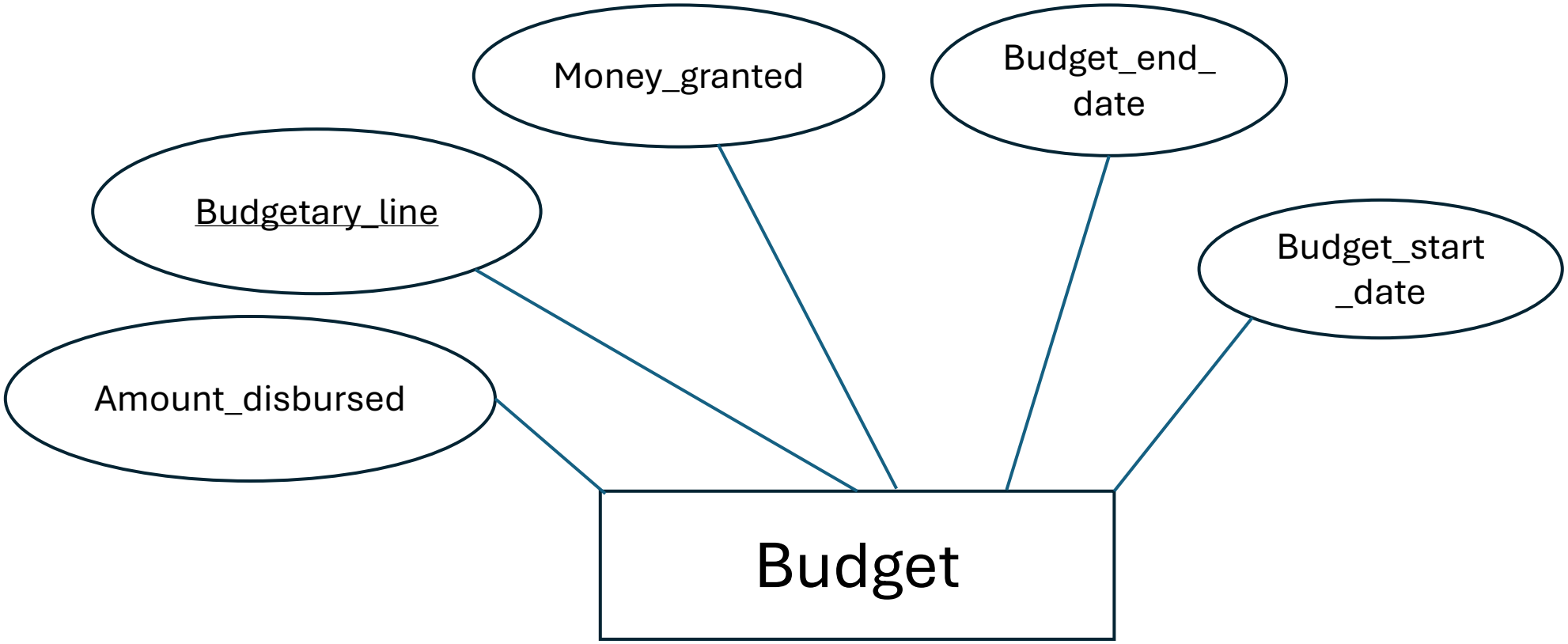
“Student” Inherits from Person :
First_name , last_name, p_Id, Affiliation, Approuval, Email, End, Start.

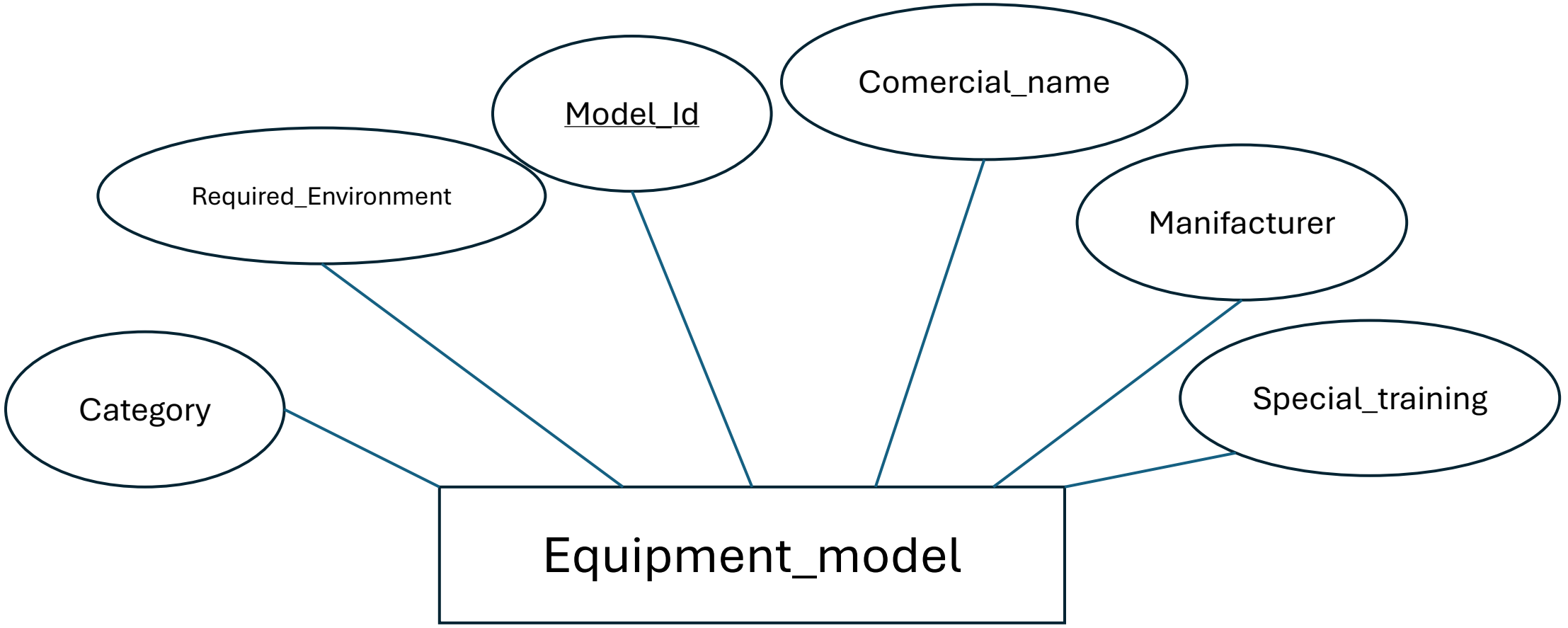


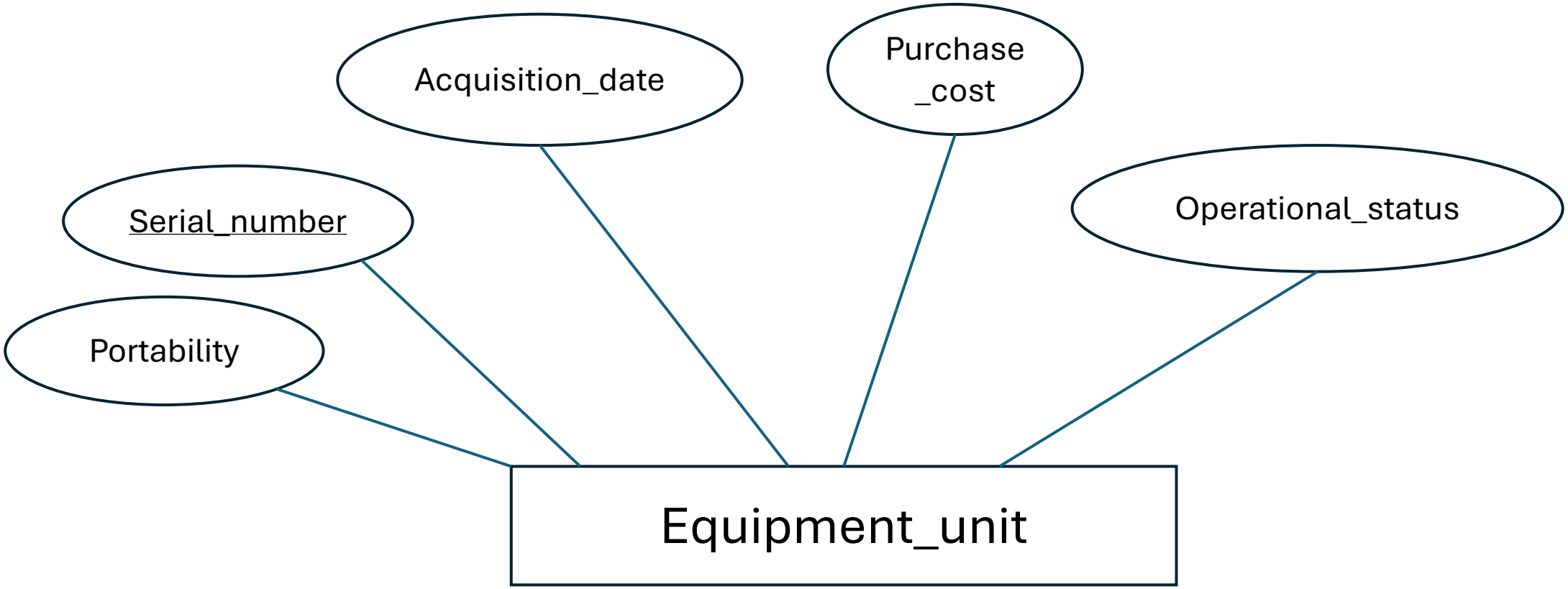
“Employee” Inherits from Person :
First_name , last_name, p_Id, Affiliation, Approuval, Email, End, Start.

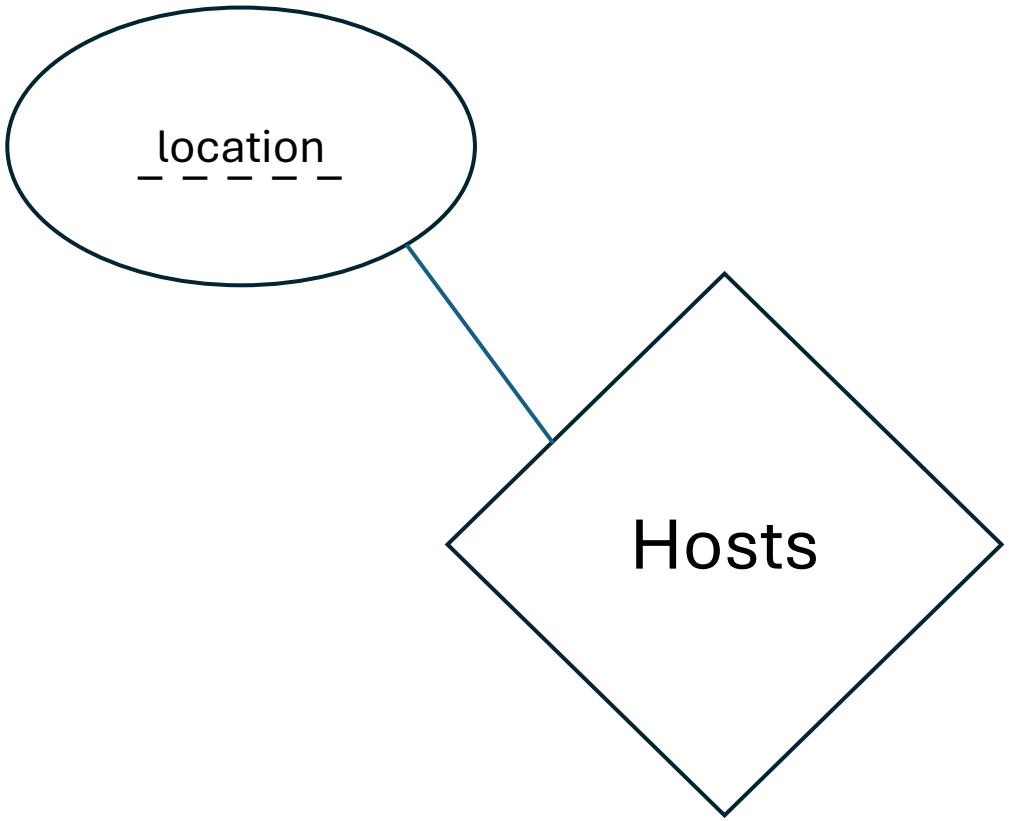

Building
Discipline
L_Id
Name
Room_number
Laboratory

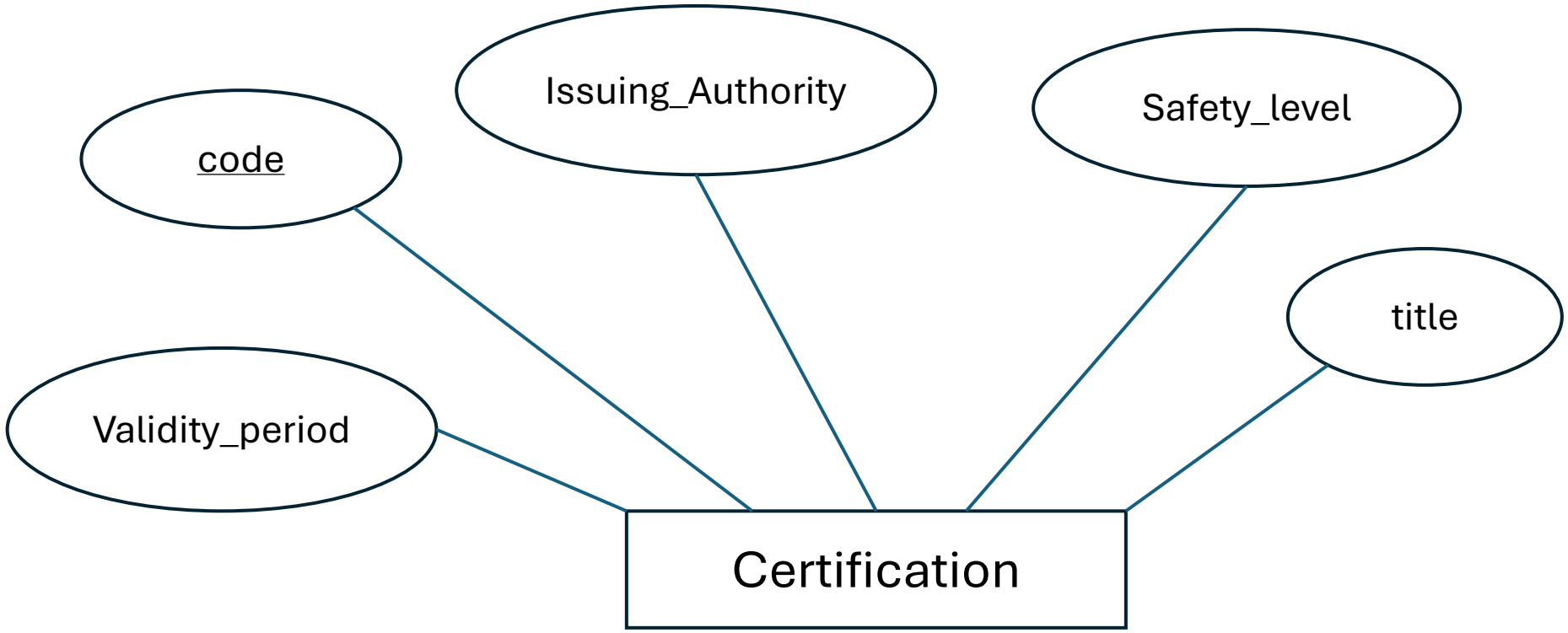

End-date
Project_code
Start-date
Status
Title
Research_project

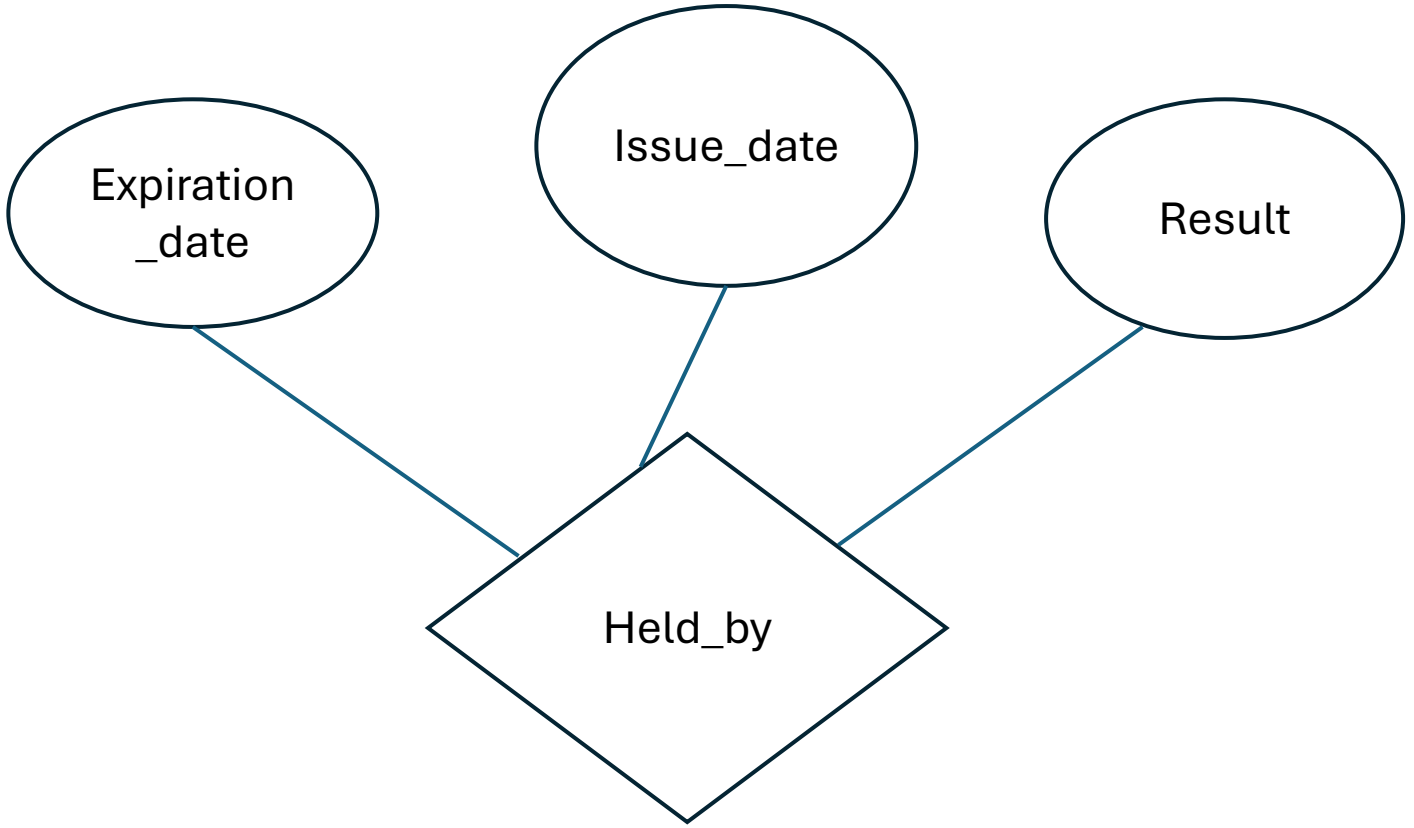

Money_granted
Budget_end_
date
Budgetary_line
Budget_start
_date
Amount_disbursed
Budget

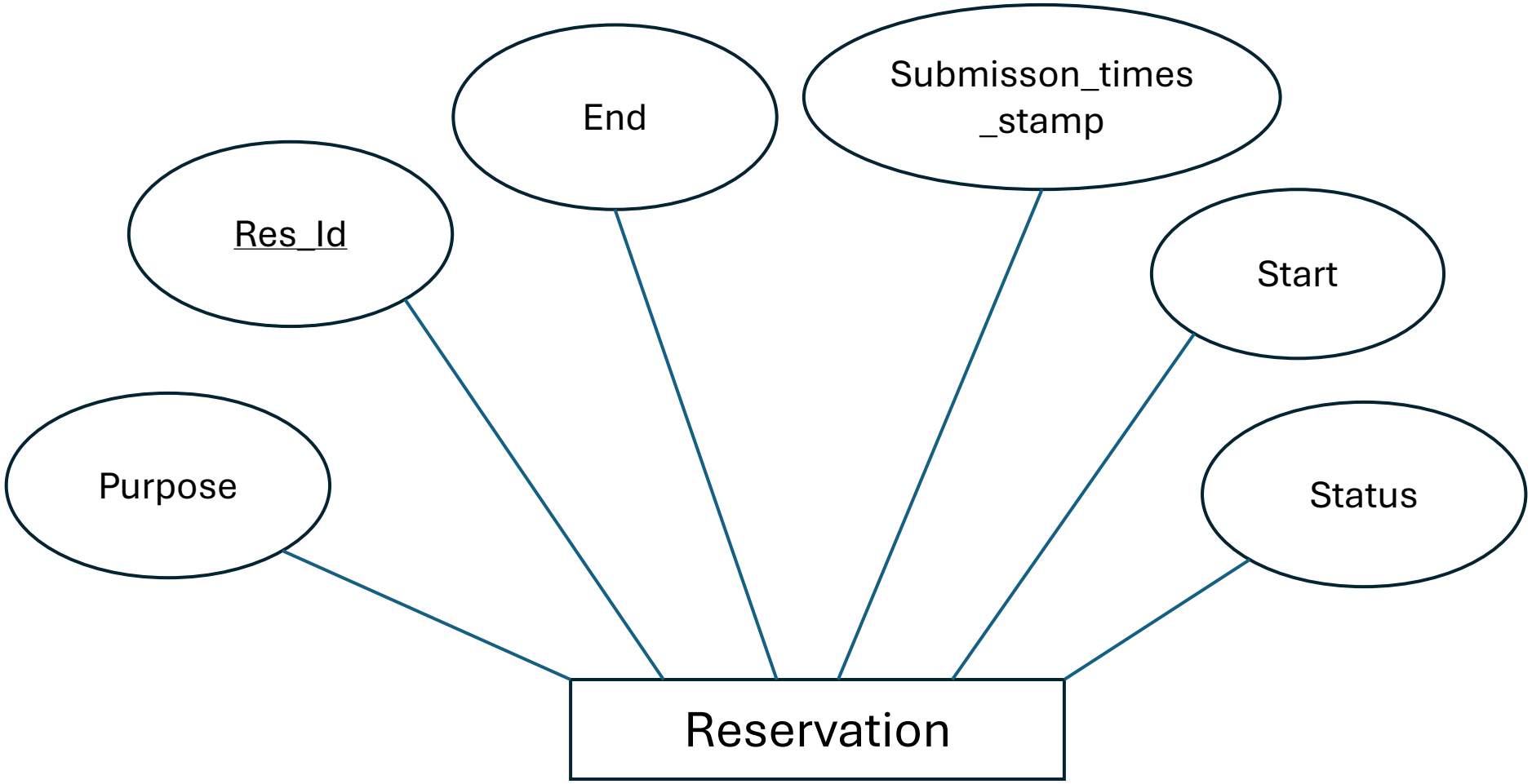

Model_Id
Comercial_name
Required_Environment
Manifacturer
Category
Special_training
Equipment_model

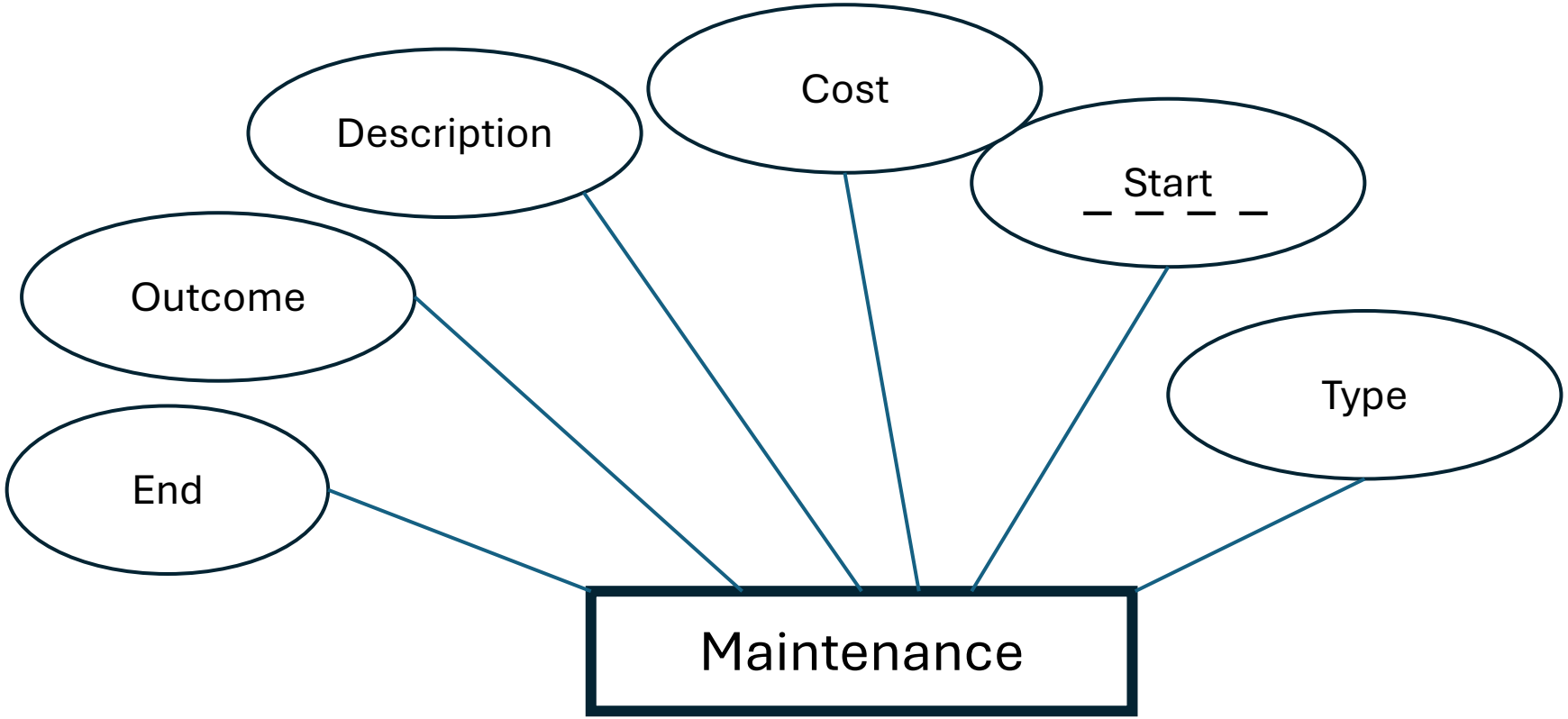

Acquisition_date
Purchase
_cost
Serial_number
Operational_status
Portability
Equipment_unit

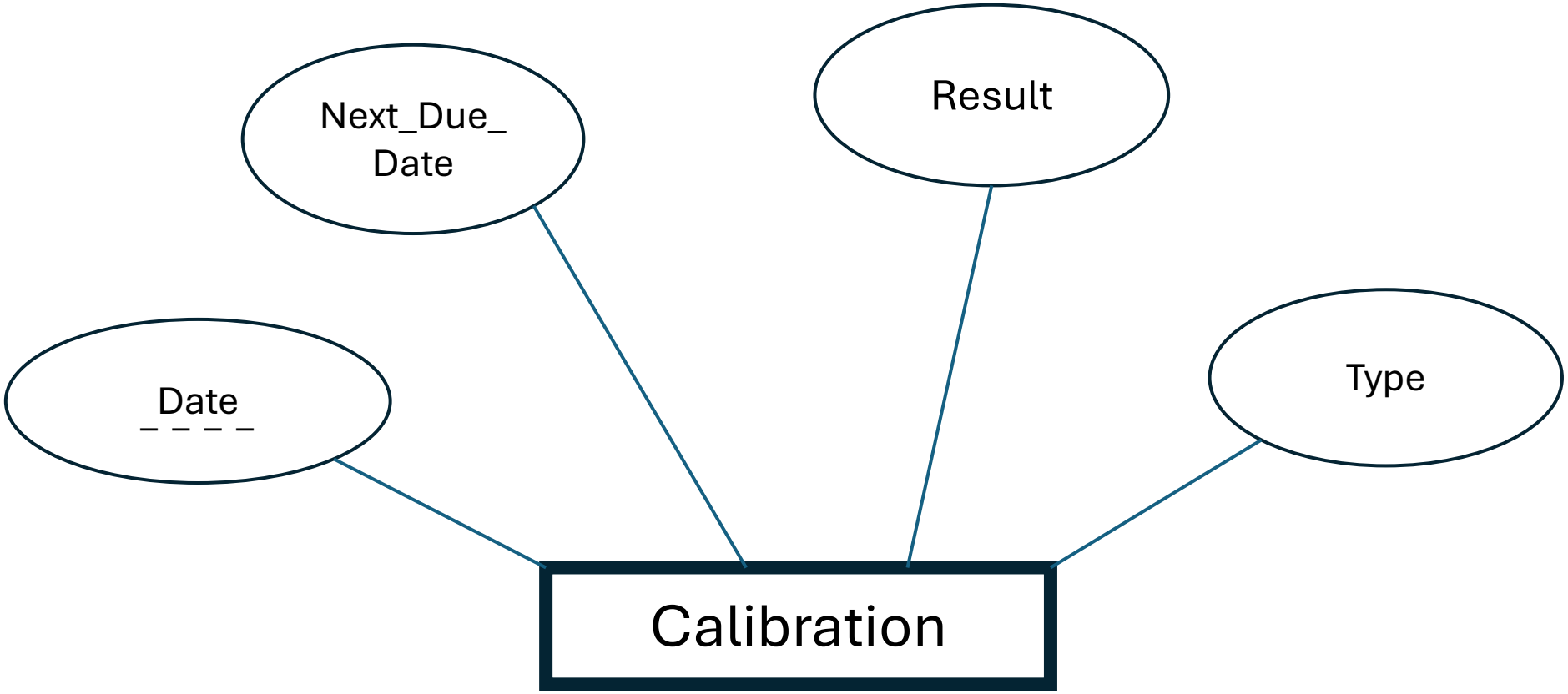

location
Hosts

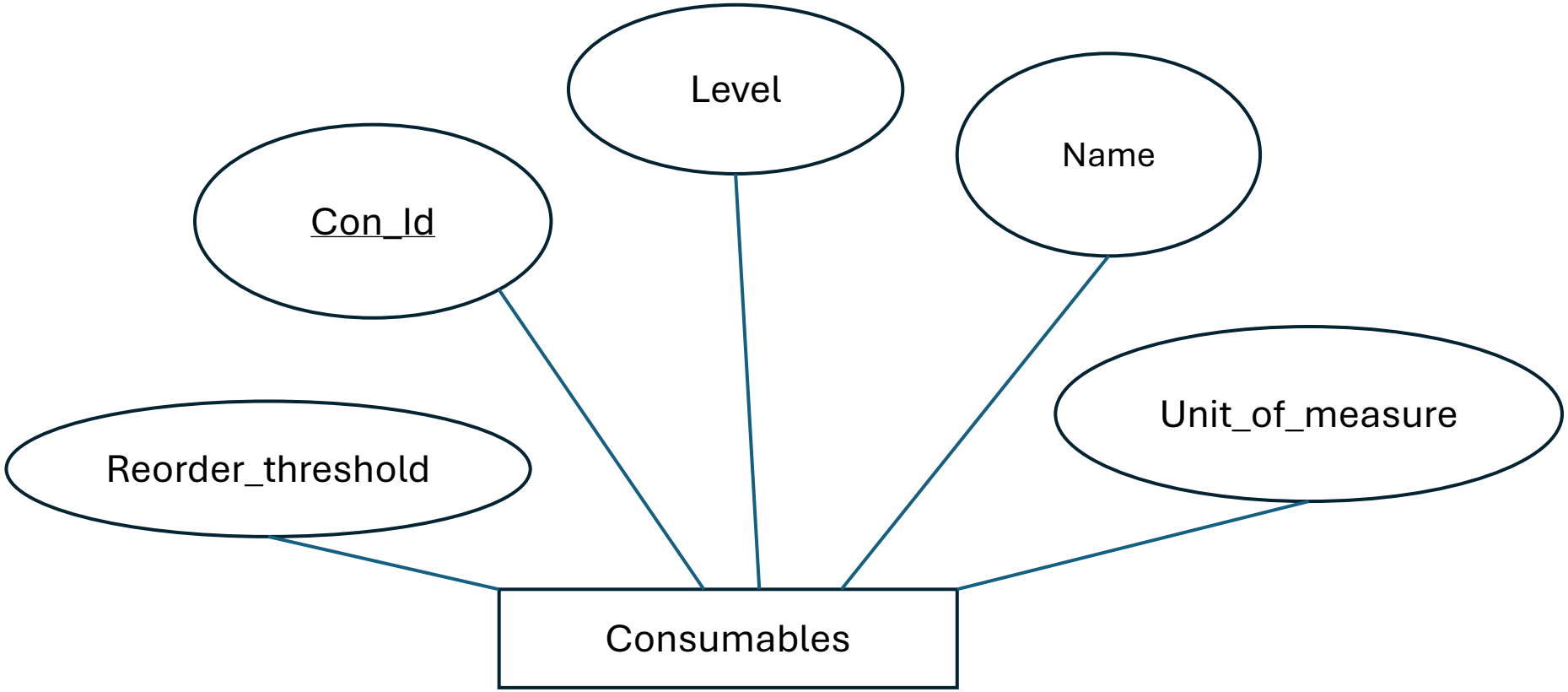

code
Issuing_Authority
Safety_level
title
Validity_period
Certification

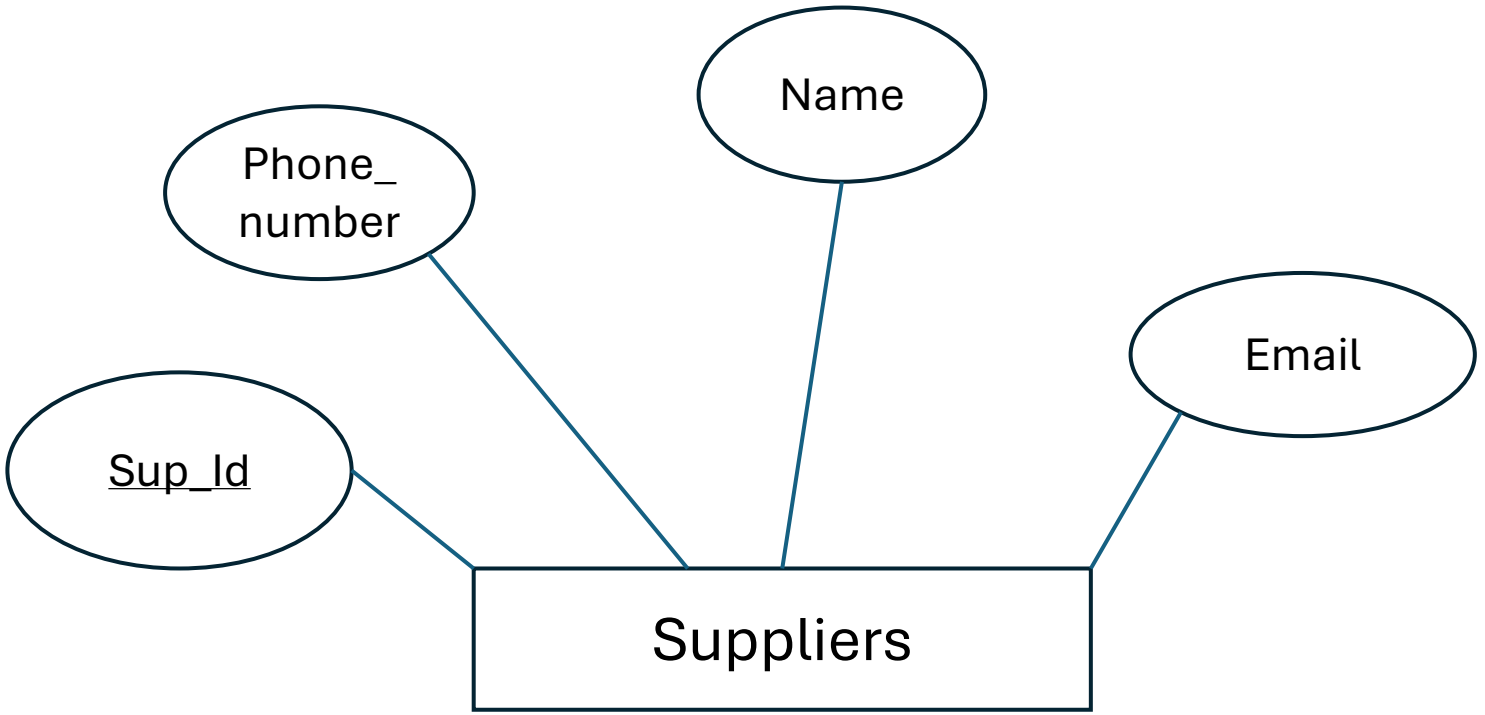

Expiration
_date
Issue_date
Result
Held_by

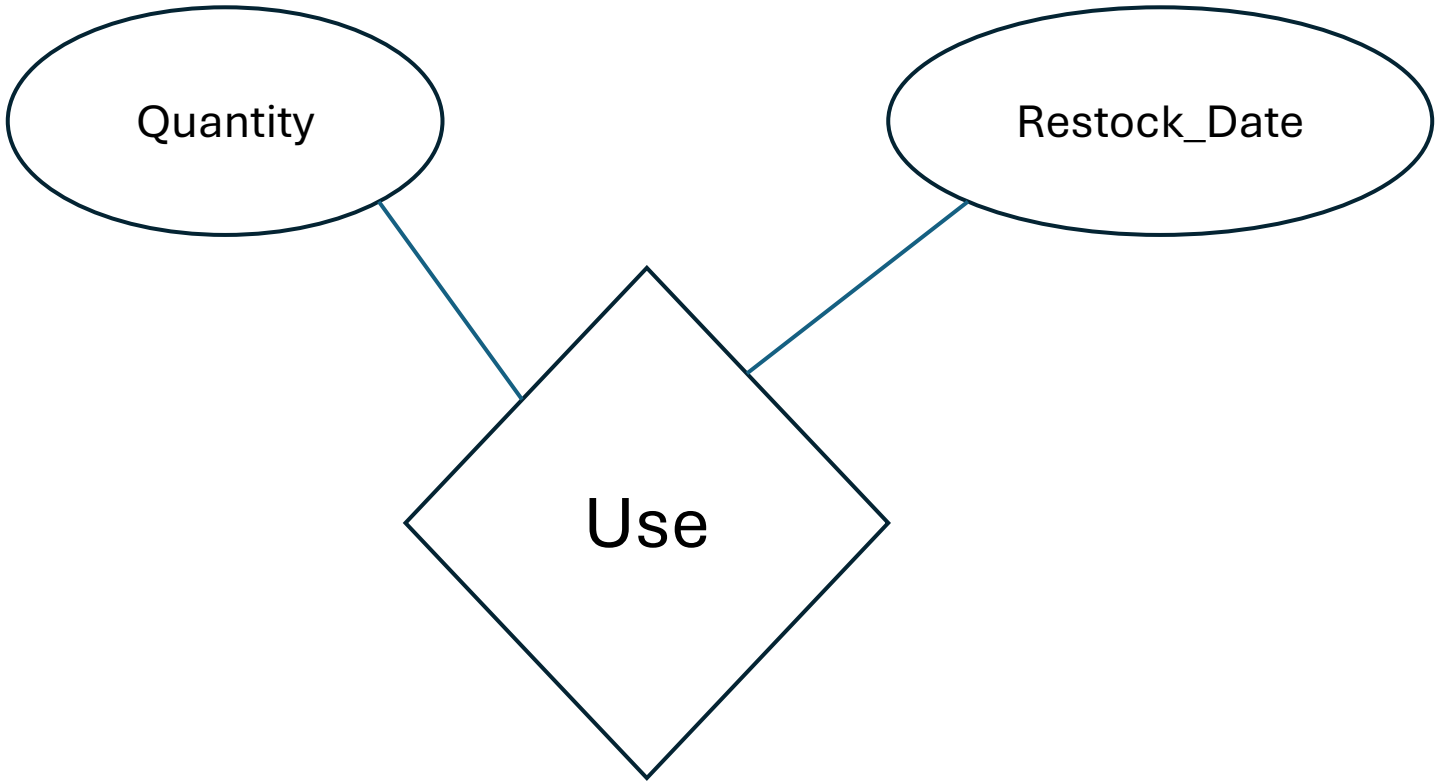

End
Submisson_times
_stamp
Res_Id
Start
Purpose
Status
Reservation

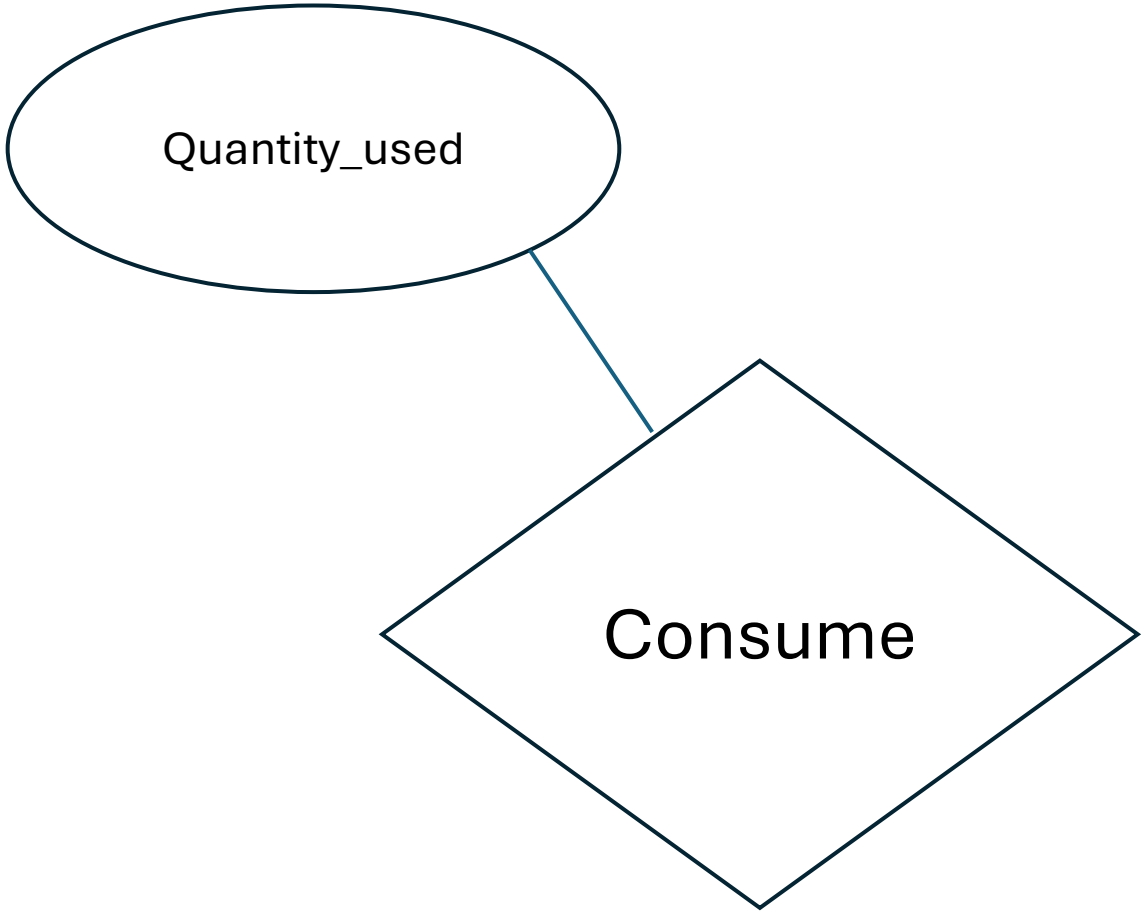

Description
Cost
Start
Outcome
Type
End
Maintenance


Next_Due_
Date
Result
Date
Type
Calibration


Level
Name
Con_Id
Unit_of_measure
Reorder_threshold
Consumables


Name
Phone_
number
Email
Sup_Id
Suppliers


Quantity
Restock_Date
Use


Quantity_used
Consume